Research Questions:

I want to answer the following questions

1. Is it more convenient for Langara students to carry their bags in front rather than on their back?

2. Does carrying the bag in front translate to a faster commute time, depending on the means of commute?

3. What kinds of bags are most convenient to carry, depending on the kind of commute?

Backstory:

I tried carrying my backpack on my front one afternoon to see how it felt on my commute. Walking was easier. It was easier to sit down on th ebench and take out my compass card from my bag's pouch. I could look down at my steps, hold the yellow rails, and move calmly to the back of the bus. I sat down without taking the bag off, and it was easy to set it on the floor. After a while, though, my back and shoulders started to ache. The strain was different from a back carry, and less covenient. That made me wonder whether everyone would have the same experience. People differ in sex, age, height, and other ways that could affect how they carry a load. So I asked: among Langara students, what percentage feel like front carry is more convenient than back carry? What relationship do the weights of Langara's students bag have with their commute time. Depending on the mode of commuting, walking, riding the bus, cycling, taking an uber, can carrying your bag a specific way affect how much time they spend on their commute? Students carry different types of bags, eg totes and crossbags. what is the percentage of students that find each kind of bag convenient to carry based off their covenience rating.

Prompt

Instruction
Generate a synthetic dataset of 1,000 students drawn from the population of students enrolled at Langara College in Fall 2026. A dataset with 1,000 rows, one per student, a header row with the variable labels. output format is csv.

Variables and labels
1 Student ID Unique Identifier - Format it consistently

1 Gender - Categorical - Male, Female
2 Age - Numeric - Years (17–60)
3 Height - Numeric - Feet
4 Student status - Categorical - Domestic, International
5 Commute length - Numeric - Km Distnace
6 Commute mode - Categorical - Public transit, Uber/rideshare, Walking, Cycling
7 Commute time - Numeric - Minutes (actual, door to door)
8 Bag weight - Numeric - Kilograms (1–10)
9 Bag type - Categorical - Backpack, Sling, Tote, Crossbody, Other
10 Carry style - Categorical - Back, Front, Crossbody, Other
11 Discomfort - Numeric - 1 (none) to 5 (severe)
12 Convenience rating - Numeric - 1 (very inconvenient) to 5 (very convenient)
13 Trip or fall during commute - Categorical - Yes, No
14 Prefer front carry - yes/no

Pay attention to relastionships between variables and simulate the dataset to be as real as possible.
Return output in CSV

Response

Coding and executing

Your downloadable CSV file is ready:

Langara Fall 2026 Synthetic Student Dataset (1,000 records)

The dataset contains:

- 1,000 synthetic student records
- All requested variables
- Realistic relationships between commute patterns, bag weight, discomfort, convenience, student status, and transport mode
- CSV format with a header row and one row per student

This dataset is synthetic and statistically simulated; it does not contain any real student information.

Analytical Questions.

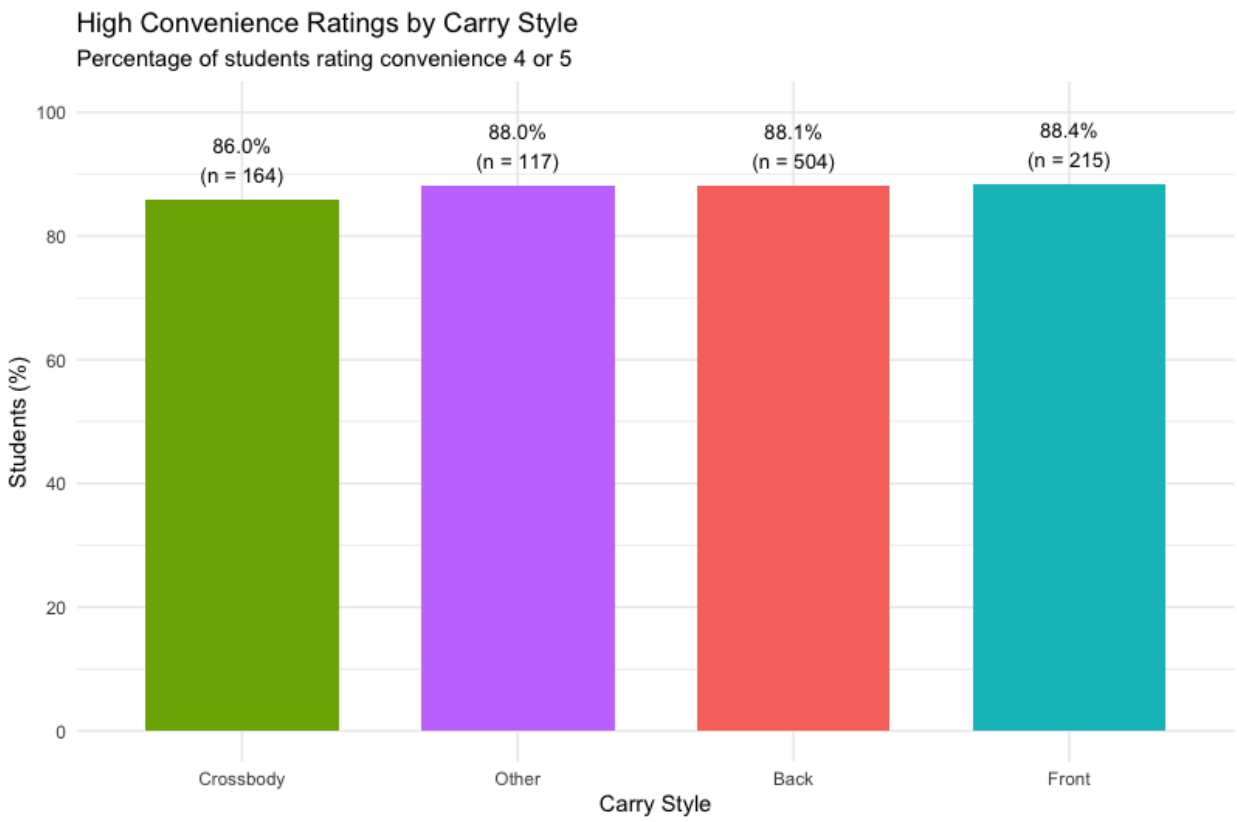
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Answer according to dataset

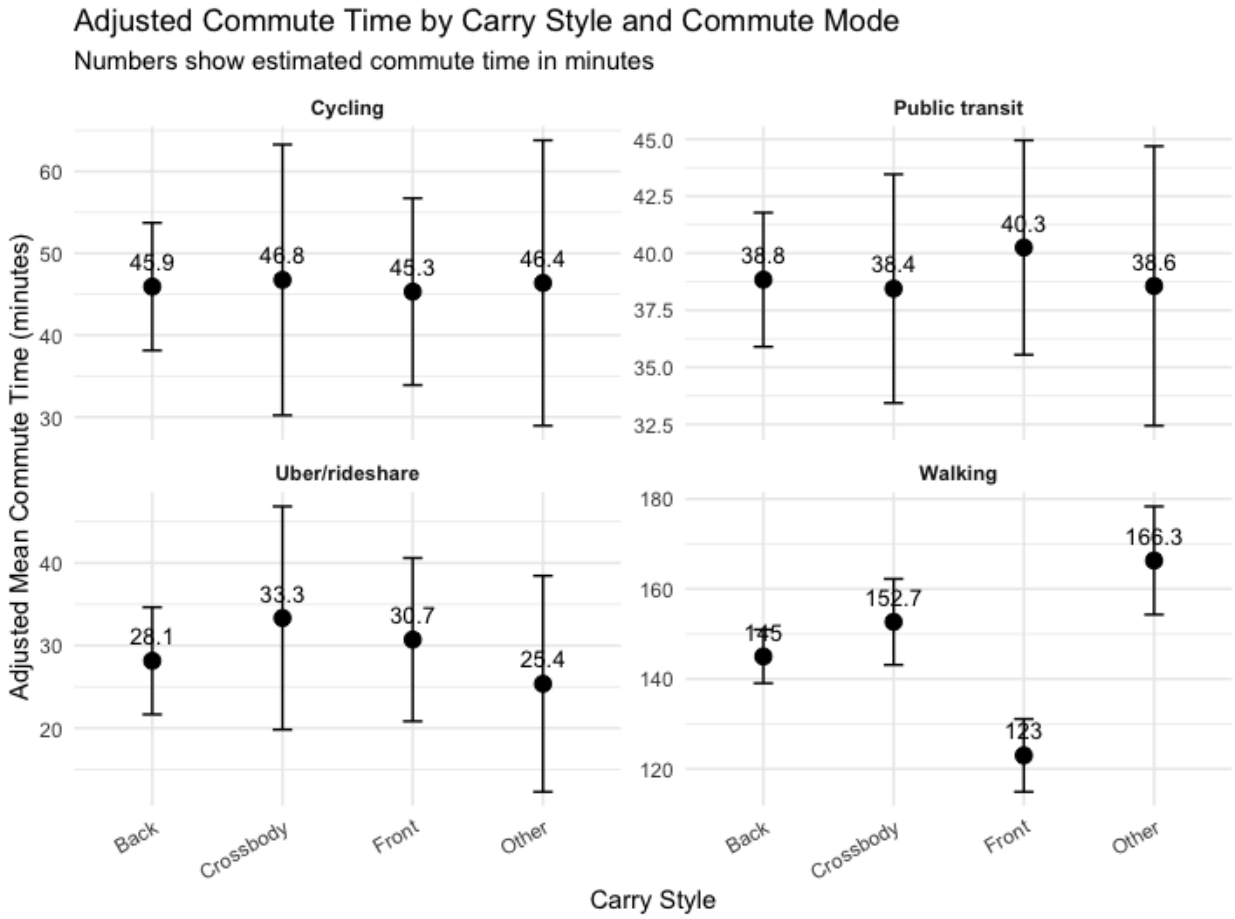
1.



Students using **Front carry** reported the highest average convenience ratings at **88.4%**, followed closely by **Back carry** at **88.1%** and **Other carry style** at **88%** while **Crossbody** reported the lowest percentage at **86%.**

The observred differences are relatively small further analysis would be needed before naming a superior carry style

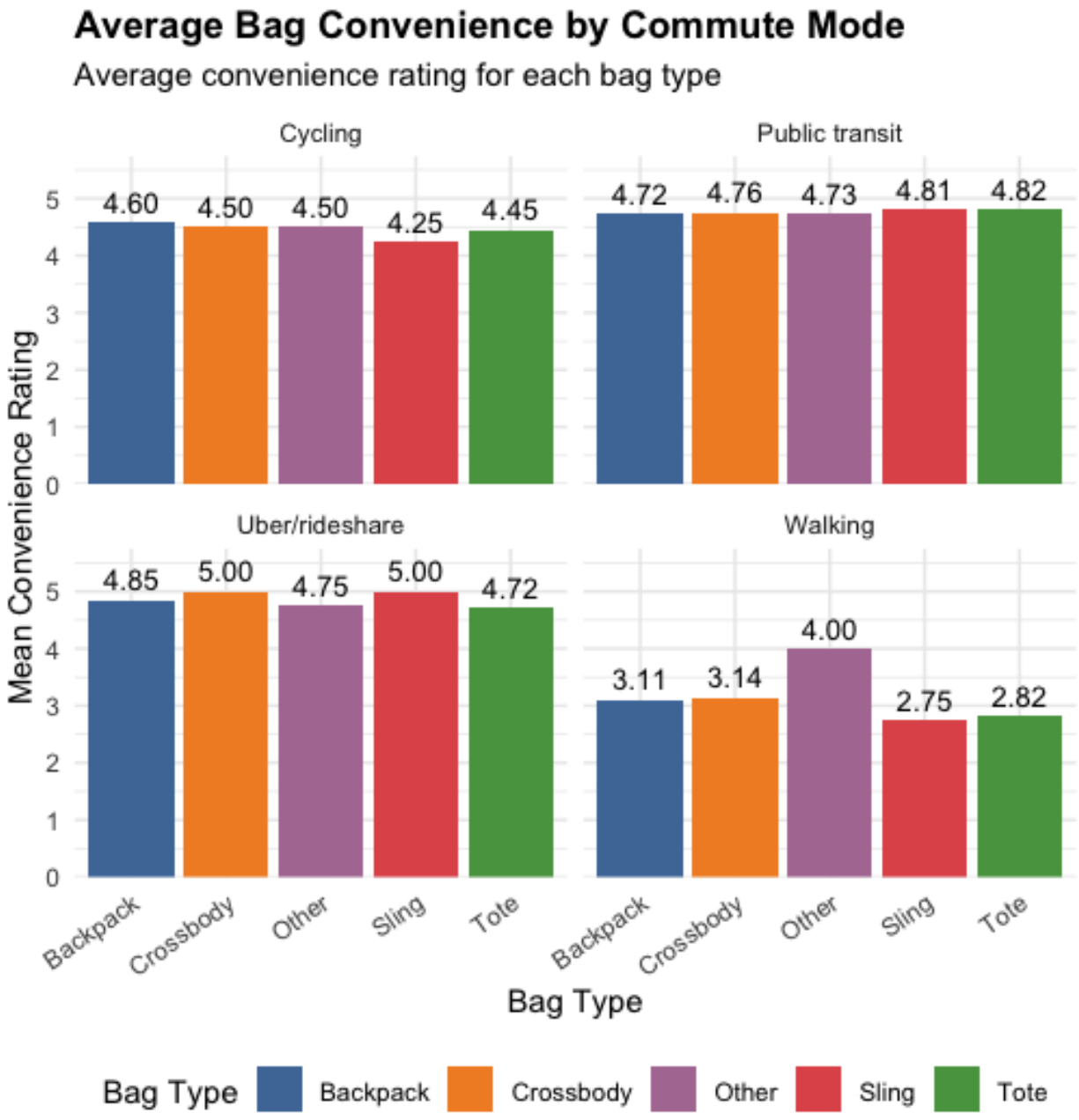
2.



Compared with front carry, back carry was associated with a adjusted mean commute time approximately 22.01 minutes longer when walking, while crossbody carry was associated with an adjusted mean approximately 29.68 minutes longer. The Other carry-style category had an adjusted mean approximately 43.33 minutes longer than front carry.

These findings suggest that the association between carry style and commute time varies by commuting method. Front carry was associated with shorter adjusted commute times than back and crossbody carry among walkers, but it was not the fastest option compared with other carry styles in the other modes of commuting.

3.



The analysis compared average convenience ratings for five bag types across four commute modes. Backpacks received the highest average rating among cyclists, totes among public transit users, crossbody and sling bags tied among Uber/rideshare users, and the "Other" category among walkers. These preliminary results suggest that the most convenient bag type may vary by commute mode.